

Time Cost of Digital Privacy Protection and Data-Driven Innovation

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First Version: March 2024

This Version: December 2025

Abstract

This paper introduces the time cost of digital privacy protection as a proxy for the instrumental value of digital privacy and analyzes its implications for consumption-data provision and data-driven innovation in a simple Schumpeterian endogenous growth model. A higher marginal time cost of privacy protection discourages data provision, raises the cost of data-driven innovation, and reduces the long-run growth rate. We show that when this marginal time cost is above (below) a threshold value, the market equilibrium features socially excessive (insufficient) innovation, due to an intertemporal business-stealing effect. This threshold depends on the innovation step size, monopoly markups in product markets, and the data share of R&D. We characterize the policy instruments that restore the social optimum. We also analyze how technological and regulatory improvements affect the marginal time cost of digital privacy protection through consumers' digital-literacy input and the intermediary's data-security input, and assess the resulting implications for innovation and long-run growth via the data-cost channel.

Keywords: data-driven innovation, data non-rivalry, data security, digital literacy, digital privacy protection, endogenous growth

JEL Classification: D83, E22, H21, O31, O41

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I thank Jürgen von Hagen and Matthias Blonski for their insightful comments during my visits at the University of Bonn and Goethe University Frankfurt. I appreciate the seminar participants at Osnabrück University and the University of Konstanz for valuable suggestions. Financial support from the University of Auckland is sincerely acknowledged.

1 Introduction

Technological progress over the past decades has dramatically reduced the marginal costs of collecting, storing, and transmitting large volume of data. The recent emergence of multimodal large language models and related artificial intelligence technologies has further expanded the scope of consumer data that can be processed at scale, including text, images, audio, and granular behavioral traces. At the same time, advances in data analytics have improved firms' ability to transform these data into operational and strategic insights. Together, these developments have turned data into a key input in firm activities, enhancing forecasting accuracy (Farboodi et al., 2019; Farboodi and Veldkamp, forthcoming), boosting productivity (Jones and Tonetti, 2020), and promoting innovation (Cong, Xie, and Zhang, 2021; Cong et al., 2022).

The pervasive use of consumer data has raised public concerns over digital privacy. The economics literature distinguishes between the intrinsic and the instrumental views of privacy (Lin, 2022; Goldfarb and Que, 2023). Under the intrinsic view, privacy is a fundamental human right, i.e., "the right to be let alone" (Warren and Brandeis, 1890), independent of downstream economic consequences. People prefer not to be observed or monitored and value privacy for reasons of dignity, autonomy, and personal control. In contrast, the instrumental view evaluates privacy in terms of its economic consequences rather than for its own sake. Privacy protects individuals from downstream harms that arise from the observation or exploitation of personal information, including personalized pricing, identity theft and data-breach risk, reputational damage, and manipulation. In short, the intrinsic view treats privacy as a "final good" that directly enters utility and affects welfare, while the instrumental view treats privacy as an "intermediate good" that affects welfare indirectly by shaping prices and market outcomes.

The current literature that features consumer data as an input into production or innovation (Jones and Tonetti, 2020; Cong, Xie, and Zhang, 2021; Cong et al., 2022; Chang, Wu, and Zhang, 2024; Wei and Xie, 2024; Chang et al., 2025; Huang, Zhang, and Dmitriev, 2025) commonly adopts an intrinsic view of digital privacy. In these frameworks, consumption data are generated as a by-product of economic activity, and data sharing is assumed to entail zero economic cost. Privacy concerns associated with the sharing of consumption data enter the consumers' preferences as a disutility term, with the associated intensity reflecting values, beliefs, or normative attitudes toward privacy, rather than economic costs linked to downstream risks or market outcomes.

Our paper contributes to the current literature by making the first attempt to incorporate the instrumental view of digital privacy concerns into a data-driven endogenous growth model. The use of online digital services typically requires consumers to disclose personal information, such as postal addresses and payment details, while their online interactions and purchasing activities reveal a more extensive and detailed set of information regarding preferences, behavioral patterns, life-cycle characteristics, and social networks. When such data are misused or breached, consumers incur time (and monetary) costs associated with damage control, account monitoring, and addressing subsequent consequences. Therefore, when deciding how much data to share, consumers face a trade-off between immediate, front-loaded benefits (such as discounts, access to free digital services, or personalized offerings) and the opportunity cost of time devoted to privacy-

protective efforts, cautious online behaviors, and managing data-breach risks.

Motivated by the seminal work of Becker (1965), we use the time cost of digital privacy protection (DPP, hereafter) as a proxy for the instrumental value of digital privacy, capturing the opportunity cost that households face when they share consumption data. Intuitively, a high marginal time cost of DPP reduces households' incentives to share data, thereby raising the cost of data-driven innovation and reducing long-run economic growth. We find that, when this marginal time cost is above (below) a threshold value, the market equilibrium features socially excessive (insufficient) innovation, due to an intertemporal business-stealing effect. This threshold depends on the innovation step size, monopoly markups in product markets, and the data share of R&D. We then characterize patent and tax policies that restore social optimum in the market equilibrium.

The key advantage of adopting the instrumental view of digital privacy is that it allows us to analyze how technological and regulatory changes affect the marginal time cost of DPP through endogenous adjustments in consumers' digital literacy effort and intermediaries' data-security inputs.¹ We then investigate the implications of these changes for innovation and long-run growth through the data-cost channel.

Intuitively, digital privacy concerns may arise in part from households' limited ability to manage their digital footprint and assess the risks associated with online activities (Acquisti, Taylor, and Wagman, 2016), intermediaries' insufficient data-security inputs (Laube and Böhme, 2016; de Cornière and Taylor, forthcoming), and weak regulatory or institutional frameworks governing data breaches (Johnson, 2024). Technological and regulatory improvements can raise the effectiveness of households' digital literacy effort and the intermediary's data-security inputs (Romanosky, Telang, and Acquisti, 2011; Marreiros et al., 2017; Godinho de Matos and Adjerid, 2022; Aridor, Che, and Salz, 2023; Fainmesser, Galeotti, and Momot, 2023; Bains and Gaidosch, 2025). To capture these factors, we endogenize the households' digital-literacy input and the intermediary's data-security input, which constitute the channels through which technological and regulatory improvements affect the data costs of innovation. We find that households' and the intermediary's decisions may exhibit strategic substitutability, depending on how such improvements shape the relative effectiveness of each input and how they jointly affect the marginal time cost of DPP.

To summarize, our paper highlights the time cost of DPP as a channel through which technological and regulatory conditions affect consumption-data provision, the cost of data-driven innovation, and long-run economic growth. Building on this finding, we propose the time cost of DPP as an objective, outcome-based measure of digital privacy concerns that can be incorporated into consumer survey design. Unlike the attitude-based measures commonly used in the literature, our approach directly captures the economic costs associated with data sharing,² and provides an objective indicator of digital privacy concerns that is comparable across countries.

Although our model focuses on consumption data sharing and digital privacy concerns, its core intuition extends naturally to the sharing and commercialization of production data among firms.

¹By contrast, the intrinsic view treats privacy-related values, beliefs, attitudes, and social norms as components of preferences that are stable or slow-moving, and may adjust only gradually to changes in the economic environment.

²Such measures also facilitate the estimation of appropriate compensation schemes and legal remedies in the event of data leakages or breaches.

As shown by Jones and Tonetti (2020), firms face business-stealing risks when disclosing proprietary production data to third parties. Advances in data-security-enhancing technologies mitigate the risks of data breaches and the leakage of business secrets (Bains and Gaidosch, 2025; Fedyk et al., 2025), thereby strengthening firms' incentives to share or sell production data. From this perspective, incorporating firms' data-security expenditures into firm-level survey design offers an objective and economically grounded measure of the indirect costs of data-driven innovation, complementing existing approaches that focus primarily on direct R&D expenditures.

The rest of the paper is organized as follows. Section 2 presents the baseline model with an exogenous marginal time cost of DPP and derives the growth rates along the balanced growth path (BGP, hereafter) under the market equilibrium and the social planner's allocation. Section 3 identifies the key factors driving differences in BGP growth rates across the two allocations and characterizes the policy instruments that restore the social optimum in the market equilibrium. Section 4 endogenizes the marginal time cost of DPP by allowing households to allocate time or resources to improving digital literacy and the data intermediary to hire labor to enhance data security. We analyze how relevant factors shape households' and the intermediary's decisions and discuss the resulting implications for the data cost for innovation. Section 5 concludes with final remarks. The Appendix contains technical proofs and extensions.

2 The Model

We feature two key elements in a Schumpeterian growth model. First, consumption data are generated as a by-product of economic activities, which are used as an input for R&D. Second, when selling consumption data, households need to devote time protecting their digital privacy.

2.1 Households

Consider an economy with a continuum of identical households. Each household is endowed with one unit of labor at every point in time, and the total population is constant at L . The representative household derives utility from consuming the final good, with the preferences given by,

$$U \equiv \int_0^\infty e^{-\rho t} \ln c_t dt, \quad (1)$$

where c_t denotes consumption at time t and ρ is the rate of time preference. Consumption activities generate κc_t units of data as a by-product, where $\kappa > 0$ captures the degree of digitization. A data intermediary collects consumption data at zero marginal cost and makes them freely accessible to the household. The household then decides how much data to sell to the intermediary, denoted by $d_t \leq \kappa c_t$, at the market price $p_{s,t}$.³ Define $\phi_t \equiv \frac{d_t}{\kappa c_t} \in [0, 1]$ as the sold data portion.

When selling consumption data, households must allocate time to DPP, including preventive actions, identifying risks in online activities, and addressing the consequences of data breaches.

³Jones and Tonetti (2020) study alternative data ownership regimes and show that assigning ownership of consumption data to households shapes data-sharing incentives and can improve welfare. We assume that households own their consumption data and therefore choose how much to sell to the data intermediary.

This time cost is given by $\frac{d_t}{\eta Z_t}$, implying a marginal time cost of DPP, $\frac{1}{\eta Z_t}$. Z_t denotes the aggregate technology level to be defined in Subsection 2.6. A higher Z_t enables households to adopt more advanced privacy-protecting technology, thereby lowering the marginal time cost of DPP.

Concerns over data privacy arise in part from households' limited ability to manage their digital footprint, insufficient data-security protection by data intermediaries, and weak regulatory and institutional frameworks governing data breaches. In this section, we treat η as a reduced-form parameter capturing the household DPP efficiency.⁴ Given Z_t , a higher η implies greater DPP efficiency and therefore a lower marginal time cost of data sales for households.

After accounting for time spent on DPP, the household supplies $1 - \frac{d_t}{\eta Z_t}$ units of labor to the market at the wage rate w_t . The household holds assets a_t , which represent claims on mutual funds owning the equity of monopolistic firms, as described below, and earns the interest rate r_t . The household chooses consumption, data sales, and asset holdings to maximize its preferences subject to the flow-of-funds constraint,⁵

$$\dot{a}_t = r_t a_t + w_t \left(1 - \frac{d_t}{\eta Z_t}\right) - c_t + p_{s,t} d_t = r_t a_t + w_t - c_t + d_t \left(p_{s,t} - \frac{w_t}{\eta Z_t}\right). \quad (2)$$

In equilibrium, the marginal cost of DPP equals the price of consumption data,⁶

$$\frac{w_t}{\eta Z_t} = p_{s,t}. \quad (3)$$

According to the Euler equation, the growth rate of consumption depends positively on the interest rate and negatively on the time preference rate,

$$\frac{\dot{c}_t}{c_t} = r_t - \rho. \quad (4)$$

2.2 Final Goods

Competitive firms combine N differentiated intermediate goods to produce the final good using a Cobb–Douglas aggregator. Profit maximization yields the market demand for each intermediate good. Let $x_{i,t}$ and $p_{i,t}$ denote, respectively, the quantity and price of intermediate good $i \in [0, N]$.

$$y_t = N \exp \left(\frac{1}{N} \int_0^N \ln x_{i,t} di \right), \quad \text{and} \quad x_{i,t} = \frac{y_t}{p_{i,t}}. \quad (5)$$

2.3 Intermediate Goods

In each intermediate-good industry $i \in [0, N]$, the industry leader hires $m_{i,t}$ units of labor to produce

$$x_{i,t} = z^{q_{i,t}} m_{i,t}, \quad (6)$$

⁴Higher household digital literacy, more effective data-security inputs by data platforms and intermediaries, and stronger data-privacy regulations and institutions reduce the time required to safeguard personal information, thereby increasing the DPP efficiency. Section 4 endogenizes the DPP efficiency, η , as a function of households' effort in improving digital literacy and the intermediary's input into data security.

⁵We restrict attention to parameter values for which the equilibrium sold data portion satisfies $\phi_t \equiv \frac{d_t}{\eta c_t} \in [0, 1]$.

⁶Because the time cost of DPP is linear in the quantity of data sold, the household's supply of consumption data is perfectly elastic. This property is key to the model tractability along the BGP.

where $z > 1$ denotes the quality step size and $q_{i,t}$ denotes the number of quality improvements that have occurred in industry i up to time t . Given its productivity level $z^{q_{i,t}}$, the industry leader acts as a monopolist and sets a price markup $\mu \in (1, z]$ over marginal cost.⁷

$$p_{i,t} = \mu \frac{w_t}{z^{q_{i,t}}}. \quad (7)$$

Combine equations (5)-(7) to get the wage payment and the monopoly profit in industry i ,

$$w_t m_{i,t} = \frac{1}{\mu} x_{i,t} p_{i,t} = \frac{1}{\mu} \frac{y_t}{N}, \quad \text{and} \quad \pi_{i,t} = x_{i,t} p_{i,t} - w_t m_{i,t} = \left(1 - \frac{1}{\mu}\right) \frac{y_t}{N}. \quad (8)$$

Monopoly profits are identical across all intermediate-good industries, and so is the labor input for production, given by $m_{i,t} = \frac{y_t}{\mu N w_t}$. Let $\pi_{i,t}$ denote monopoly profit in industry i . Total monopoly profits are $\pi_t \equiv N \pi_{i,t}$, and total labor input for intermediate-good production is $m_t \equiv N m_{i,t}$.

2.4 Data Intermediary

Due to data nonrivalry, the data intermediary industry cannot be modeled as perfectly competitive. Following Jones and Tonetti (2020), we assume that the data intermediary operates as a monopolist, subject to free entry at a vanishingly small cost. The intermediary aggregates all purchased consumption data into a single bundle, $D_t = d_t L$, and sells identical copies of this bundle to N innovators in the intermediate-good industries at price $p_{b,t}$. Competition from potential entrants drives profits to zero in equilibrium.⁸

$$p_{s,t} d_t L = N p_{b,t} D_t, \quad \Rightarrow \quad p_{b,t} = \frac{p_{s,t}}{N} = \frac{w_t}{N \eta Z_t}. \quad (9)$$

where N reflects the nonrivalry of the data bundle and $\frac{1}{\eta Z_t}$ the marginal time cost of DPP.

2.5 R&D and Innovation

As shown above, monopoly profits are equalized across intermediate-good industries, and so are the patent values of successful innovations. Let $v_{i,t}$ and v_t denote the patent value in industry i and the aggregate patent value, respectively, with $v_t \equiv N v_{i,t}$. Let λ_t denote the average arrival rate of innovation across industries. At time t , mutual funds that own the equity of all intermediate-good firms receive total monopoly profits and capital gains, while facing expected losses due to

⁷Grossman and Helpman (1991) assume a limit-pricing condition between the incumbent and the previous leader, which pins down the markup at the quality step size, $\mu = z$. Evans, Quigley, and Zhang (2003) and Chu and Cozzi (2019) consider a price-regulation regime in which the markup satisfies $\mu \in (1, z]$. Following Li (2001) and Iwaisako and Futagami (2013), Yu and Lai (2025) introduce a policy parameter $\beta \in (\frac{1}{z}, 1]$ capturing patent breadth and reformulate the markup as $\mu = \beta z \in (1, z]$.

⁸Because the data bundle is nonrival, the marginal cost of selling an additional copy is zero. Under the zero-profit condition, the intermediary spreads the total cost of consumption data acquisition evenly across all buyers when setting the price $p_{b,t}$. Thus, selling the data bundle to more buyers lowers the price paid by each buyer.

creative destruction when successful innovations replace incumbent technologies. Combining the no-arbitrage condition with the Euler equation (4) yields the patent value along the BGP,

$$r_t = \frac{\pi_t + \dot{v}_t - \lambda_t v_t}{v_t} = \frac{\pi_t}{v_t} + g_t - \lambda = \frac{\pi_t}{v_t} + r_t - \rho - \lambda, \Rightarrow v_t = \frac{\pi_t}{\rho + \lambda_t}. \quad (10)$$

where $g_t \equiv \frac{\dot{c}_t}{c_t} = \frac{\dot{v}_t}{v_t}$ denotes the growth rate along the BGP.

In each intermediate-good industry, an innovator hires labor $s_{i,t}$ and purchases the data bundle D_t to conduct R&D for quality improvement. The arrival rate of innovation is a Cobb–Douglas function of these two inputs,

$$\lambda_{i,t} = \psi \left(\frac{D_t}{Z_t \alpha} \right)^\alpha \left(\frac{s_{i,t}}{1 - \alpha} \right)^{1-\alpha} = \psi \left(\frac{\tilde{D}_t}{\alpha} \right)^\alpha \left(\frac{s_{i,t}}{1 - \alpha} \right)^{1-\alpha}, \text{ where } \tilde{D}_t \equiv \frac{D_t}{Z_t} \quad (11)$$

denotes the data bundle normalized by the level of aggregate technology, $\psi > 0$ denotes the innovation productivity, and $\alpha \in (0, 1]$ denotes the data share of R&D.

Innovator i chooses these two inputs to maximize its expected profit,

$$\max_{D_t, s_{i,t}} \lambda_{i,t} v_{i,t} - p_{b,t} D_t - w_t s_{i,t}, \Rightarrow \frac{D_t}{\alpha} p_{b,t} = \frac{\tilde{D}_t}{\alpha} Z_t p_{b,t} = \frac{s_{i,t}}{1 - \alpha} w_t = \lambda_{i,t} v_{i,t} \quad (12)$$

Because patent values are identical across all intermediate-good industries, innovators choose the same amount of labor and purchase the same quantity of the data bundle. Thus, the innovation arrival rate is equalized across industries. We drop the industry subscript i and use λ_t as follows.⁹

As the data bundle is nonrival, the data intermediary supplies identical copies to all innovators. This feature alters how the innovation-productivity parameter scales with the number of industries. Rather than scaling down proportionally with N , the innovation-productivity parameter scales with $N^{1-\alpha}$, because labor remains rivalrous and $1 - \alpha$ is the labor share in the R&D process, as shown in equation (13). Let $s_t \equiv N s_{i,t}$ denote the total labor input in R&D.

$$\lambda_t = \lambda_{i,t} = \psi \left(\frac{\tilde{D}_t}{\alpha} \right)^\alpha \left(\frac{\frac{s_t}{N}}{1 - \alpha} \right)^{1-\alpha} = \frac{\psi}{N^{1-\alpha}} \left(\frac{\tilde{D}_t}{\alpha} \right)^\alpha \left(\frac{s_t}{1 - \alpha} \right)^{1-\alpha}. \quad (13)$$

2.6 Aggregate Technology and Economic Growth

Define aggregate technology Z_t as follows. Combining it with equations (5) and (6), the final-good production function becomes linear in the total labor input in intermediate-good production.

$$Z_t \equiv \exp \left(\frac{\ln z}{N} \int_0^N q_{i,t} di \right) \text{ and } y_t = \exp \left(\frac{\ln z}{N} \int_0^N q_{i,t} di \right) \cdot N \exp \left(\frac{1}{N} \int_0^N \ln m_{i,t} di \right) = Z_t m_t. \quad (14)$$

⁹In the conventional Schumpeterian growth model, innovation occurs within each intermediate-good industry, and the number of industries is typically normalized to one and represented as a continuum indexed by $i \in [0, 1]$. When R&D requires multiple inputs combined through a CES aggregator, the functional form of the innovation arrival rate is identical at the industry and aggregate levels; see Chu and Cozzi (2019) for an example. By contrast, when the number of intermediate-good industries is explicitly modeled as N , the innovation-productivity parameter in the aggregate R&D function must be scaled down by a factor of N if all R&D inputs are rivalrous. This adjustment reflects the fact that R&D inputs are distributed across N industries, which reduces the arrival rate of innovation in each industry.

Use the law of large numbers to rewrite the level of aggregate technology Z_t . Differentiate the log of Z_t with respect to time to get the growth rate of aggregate technology,

$$Z_t = \exp \left(\ln z \int_0^t \lambda_i dj \right), \quad \text{and} \quad g_t \equiv \frac{\dot{Z}_t}{Z_t} = \lambda_t \ln z. \quad (15)$$

2.7 The Market Equilibrium

Although consumption data are generated as a by-product at zero cost, selling them to the intermediary requires households to devote time to DPP. Due to the linear time cost of DPP and the intermediary's zero-profit condition, the price of the normalized data bundle $\tilde{D}_t$ is proportional to the wage rate, as shown in equation (9).

$$\frac{p_{b,t} Z_t}{w_t} = \frac{1}{N\eta} \quad (16)$$

Combining this with equation (12), the innovator's demand for the normalized data bundle is proportional to its demand for labor.

$$\tilde{D}_t = \frac{\alpha s_{i,t}}{1 - \alpha} \frac{w_t}{p_{b,t} Z_t} = \frac{\alpha s_{i,t}}{1 - \alpha} N\eta, \quad (17)$$

which allows us to simplify the Cobb-Douglas innovation function as follows.

Besides hiring $s_{i,t}$ units of labor, the innovator purchases the data bundle $\tilde{D}_t$. Creating the data bundle requires $\frac{1}{\eta} \frac{\tilde{D}_t}{Z_t}$ units of time devoted to DPP at the aggregate level. Due to data nonrivalry, this time cost is essentially shared among the N innovators; see equation (9). Thus, the data bundle purchased by an individual innovator is implicitly equal to the labor input of $\frac{1}{N\eta} \frac{\tilde{D}_t}{Z_t} = \frac{\tilde{D}_t}{N\eta}$.

Define the effective time input for R&D of innovator i as

$$\mathbb{s}_{i,t} \equiv s_{i,t} + \frac{\tilde{D}_t}{N\eta} = \frac{s_{i,t}}{1 - \alpha}, \quad (18)$$

where the last equality follows from the innovator's first-order conditions (17). Combining equations (11) with (17)-(18), we rewrite the arrival rate of innovation into an equivalent representation,

$$\lambda_t = \lambda_{i,t} = \psi (N\eta)^\alpha \mathbb{s}_{i,t} = \tilde{\psi} \mathbb{s}_t, \quad \text{where} \quad \mathbb{s}_t \equiv N \mathbb{s}_{i,t} = s_t + \frac{\tilde{D}_t}{\eta} \quad \text{and} \quad \tilde{\psi} \equiv \frac{\psi}{N} (N\eta)^\alpha \quad (19)$$

denote the aggregate effective R&D time input and the augmented innovation productivity, respectively. The innovator's optimization yields its demand for the effective time input,

$$\max_{\mathbb{s}_{i,t}} \lambda_{i,t} v_{i,t} - w_t \mathbb{s}_{i,t}, \quad \Rightarrow \quad \tilde{\psi} v_{i,t} = w_t \quad (20)$$

Combine equations (8), (10), and (14) to solve for the wage rate and obtain the value of a patent along the BGP,

$$w_t = \frac{1}{\mu} \frac{y_t}{m_t} = \frac{1}{\mu} Z_t, \quad \text{and} \quad v_{i,t} = \frac{v_t}{N} = \frac{\frac{\pi_t}{N}}{\rho + \lambda_t} = \frac{(1 - \frac{1}{\mu}) Z_t \frac{m_t}{N}}{\rho + \lambda_t}. \quad (21)$$

According to the labor market clearing condition, households, after deducting the time for DPP, supply their labor to the market for R&D activities, s_t , and the production of intermediate goods, m_t . Equivalently speaking, the total time endowment L is allocated between the broadly defined R&D activities, s_t , and the production of intermediate goods, m_t .

$$s_t + m_t = L \left(1 - \frac{1}{\eta} \frac{d_t}{Z_t} \right) = L - \frac{\tilde{D}_t}{\eta}, \quad \Rightarrow \quad s_t + m_t = L. \quad (22)$$

Combine equations (19)-(22) to solve for the BGP innovation rate and the sold data portion.

$$\lambda^{me} = \frac{g^{me}}{\ln z} = \mathbf{Q} \left(1 - \frac{1}{\mu} \right) - \frac{\rho}{\mu} \quad \text{and} \quad \phi^{me} = \frac{\eta}{\kappa} \left(\frac{\mu \mathbf{Q}}{\rho + \mathbf{Q}} - 1 \right), \quad (23)$$

where $\mathbf{Q} \equiv \tilde{\psi} L = \frac{\psi}{N} (N\eta)^\alpha L$ denotes a composite parameter.

2.8 The Social Planner's Allocation

Equation (22) shows the aggregate allocation of time among production, R&D, and DPP. Creating the (normalized) data bundle $\tilde{D}_t$ requires the marginal time cost of $\frac{1}{\eta}$ at the aggregate level. As the data bundle is nonrival, its marginal time cost for each innovator is $\frac{1}{N\eta}$. The social planner chooses the quantities of the data bundle and R&D labor by equating their expected social marginal benefits to their respective social marginal costs for each innovator. Thus, the normalized data bundle is proportional to R&D labor, and the arrival rate of innovation is linear in the effective time input for R&D, mirroring the market-equilibrium outcome; see equations (17) and (19).

The social planner chooses consumption and the effective time input for R&D to maximize the household's preferences, subject to the relevant resource constraints and the technologies governing production and innovation.

$$\max_{c_t, s_t} U \equiv \int_0^\infty e^{-\rho t} \ln c_t dt, \quad \text{s.t.} \quad c_t = \frac{y_t}{L} = \frac{Z_t m_t}{L} = Z_t - \frac{Z_t s_t}{L}, \quad \lambda_t = \tilde{\psi} s_t, \quad \dot{Z}_t = Z_t \lambda_t \ln z. \quad (24)$$

The solutions to the BGP innovation rate and the shared data portion are as follows:

$$\lambda^{sp} = \frac{g^{sp}}{\ln z} = \mathbf{Q} - \frac{\rho}{\ln z}, \quad \text{and} \quad \phi^{sp} = \frac{\eta}{\kappa} \left(\frac{\mathbf{Q} \ln z}{\rho} - 1 \right). \quad (25)$$

3 Comparing the BGP Growth Rate under Two Allocations

Let us start with the case of $\mu = z$. In figure 1, the black line shows the market-equilibrium BGP innovation schedule as a linear function of the composition parameter $\mathbf{Q}$, while the green line shows the social planner's schedule. Define three threshold values of the composite parameter,¹⁰

$$\tilde{\mathbf{Q}}^{me} \equiv \frac{\rho}{\mu - 1}, \quad \tilde{\mathbf{Q}}^{sp} \equiv \frac{\rho}{\ln z}, \quad \hat{\mathbf{Q}} \equiv \rho \left(\frac{\mu}{\ln z} - 1 \right). \quad (26)$$

¹⁰We focus on the parameter region in which the sold data portion does not exceed its upper bound under each allocation, i.e., $\phi_t < 1$. From equations (23) and (25), this condition holds when the degree of digitization is sufficiently high: $\kappa \geq \underline{\kappa}^{me} \equiv \eta \left(\frac{\mu \mathbf{Q}}{\rho + \mathbf{Q}} - 1 \right)$ in the market equilibrium and $\kappa \geq \underline{\kappa}^{sp} \equiv \eta \left(\frac{\mathbf{Q} \ln z}{\rho} - 1 \right)$ in the social planner's allocation.

To ensure a non-negative arrival rate of innovation, $\lambda_t = \tilde{\psi} s_t \geq 0$, we focus on the cases where the composite parameter is no less than its respective threshold value in each allocation: $Q \geq \tilde{Q}^{me}$ in the market equilibrium and $Q \geq \tilde{Q}^{sp}$ in the social planner's allocation. The two schedules intersect at point A, where $Q = \hat{Q}$.

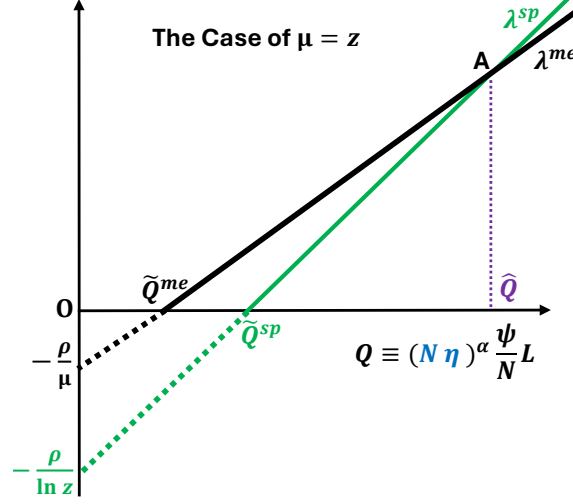


Figure 1: BGP Innovation Schedules

3.1 Social Gains from Innovation vs The Business-Stealing Effect

First, consider the social planner's problem. A one-time increase in the effective R&D time input raises the arrival rate of innovation by $\frac{\partial \lambda_t}{\partial s_t} = \tilde{\psi}$ and accelerates the growth rate of aggregate technology at time t , thereby permanently increasing future output and consumption. To evaluate the social marginal benefit, the social planner discounts the changes in future utility flows along the BGP at the rate of time preference ρ , and converts them into units of current consumption by normalizing with the marginal utility of current consumption, $1/C_t$. The associated social marginal cost is the marginal product of labor, Z_t , capturing the opportunity cost in terms of foregone current final-good output and consumption. The social planner chooses the effective R&D time input by equating the expected social marginal benefit to the social marginal cost, which pins down the socially optimal BGP innovation rate.

$$\frac{\frac{\partial \ln C_t}{\partial \ln Z_t} \frac{\partial \ln Z_t}{\partial \lambda_t} \frac{\partial \lambda_t}{\partial s_t}}{\rho \frac{1}{C_t}} = \frac{\ln z}{\rho \frac{1}{Z_t m_t}} \tilde{\psi} = Z_t \quad \Rightarrow \quad \lambda^{sp} = \tilde{\psi}(L - m_t) = Q - \frac{\rho}{\ln z}. \quad (27)$$

A larger quality step size z raises the social return to innovation, inducing the social planner to allocate more effective R&D time and thereby generating a higher growth rate.¹¹

Next, consider the innovator's decision in the market equilibrium. According to equation (20), an increase in the innovator's effective R&D time input raises the arrival rate of innovation by $\frac{\partial \lambda_{i,t}}{\partial s_{i,t}} = N \tilde{\psi}$. The innovator's marginal benefit is the resulting increase in the expected patent value,

¹¹At the aggregate level, a fraction of the effective R&D time input takes the form of R&D labor, $s_t = (1 - \alpha)s_t$, while the remainder is the time devoted to DPP required for creating the data bundle, $\frac{1}{\eta} \frac{D_t}{Z_t} = \alpha s_t$; see subsection 2.7.

$N\tilde{\psi}v_{i,t}$, while the marginal cost is the wage rate w_t . Each innovator chooses its effective R&D time input by equating the expected marginal benefit to the marginal cost, which yields the market-equilibrium BGP innovation rate.

$$v_{i,t} \frac{\partial \lambda_{i,t}}{\partial s_{i,t}} = \frac{(1 - \frac{1}{\mu}) \frac{Z_t m_t}{N}}{\rho + \lambda_t} N\tilde{\psi} = w_t = \frac{Z_t}{\mu} \Rightarrow \lambda^{me} = \tilde{\psi}(L - m_t) = \mathbf{Q}(1 - \frac{1}{\mu}) - \frac{\rho}{\mu}. \quad (28)$$

The markup ratio μ affects R&D inputs and long-run growth through two channels in our model. First, a higher μ increases monopoly profits and, consequently, the value of a patent, which raises the expected return to successful innovation and induces innovators to increase the effective R&D time input. In the market equilibrium, successful innovators become new industry leaders and capture the entire monopoly profit, reflecting the business-stealing effect. Second, a higher markup, by reducing the wage rate, $w_t = \frac{Z_t}{\mu}$, lowers the marginal cost of R&D, which further induces innovators to raise their effective R&D time input. Through these two channels, a higher markup increases the arrival rate of innovation, leading to a higher BGP growth rate.

3.2 Efficiency of Digital Privacy Protection

A higher η makes the DPP less time-intensive, reducing the opportunity cost of supplying data. Then, the lower equilibrium prices of consumption data and the data bundle induce innovators to purchase more of the data bundle. The resulting data deepening raises the augmented innovation productivity, $\tilde{\psi} \equiv \frac{\Psi}{N}(\mathbf{N}\eta)^\alpha$, thereby increasing the expected payoff to R&D and inducing innovators to raise their effective R&D time input. Thus, the DPP efficiency affects the arrival rate of innovation positively through the augmented innovation productivity and the R&D time input.

Meanwhile, a higher arrival rate of innovation increases the likelihood that a successful innovator will be displaced sooner by future entrants, thereby reducing the expected marginal benefit from R&D in the market equilibrium. The intertemporal business-stealing externality, captured by the term λ_t in the denominator of equation (28), explains why the market-equilibrium BGP innovation schedule is flatter than that under the social planner's allocation, as shown in figure 1.

Define three threshold values of η that correspond to the three threshold values of $\mathbf{Q}$.

$$\eta^{me} \equiv (\mathbf{K}\tilde{\mathbf{Q}}^{me})^{\frac{1}{\alpha}}, \quad \eta^{sp} \equiv (\mathbf{K}\tilde{\mathbf{Q}}^{sp})^{\frac{1}{\alpha}}, \quad \hat{\eta} \equiv (\mathbf{K}\hat{\mathbf{Q}})^{\frac{1}{\alpha}}, \quad \text{where } \mathbf{K} \equiv \frac{N^{1-\alpha}}{\psi L}. \quad (29)$$

Lemma 1. *If $\eta \in (\tilde{\eta}^{me}, \hat{\eta})$, the market equilibrium features socially excessive BGP innovation rate, $\lambda^{me} > \lambda^{sp}$. If $\eta > \hat{\eta}$, the opposite applies, $\lambda^{me} < \lambda^{sp}$.*

In the current model, the data share of R&D activities, α , is taken as exogenously given. Suppose instead that innovators can choose among alternative innovation technologies that differ only in the data share of R&D. If the DPP efficiency is so high that the normalized data bundle is cheaper than labor, i.e., $\frac{1}{N\eta} < 1$, innovators optimally adopt the innovation technology with a higher data share and allocate a larger fraction of R&D expenditure toward the relatively cheaper input, i.e., the data bundle. This endogenous adoption of innovation technologies reinforces the data-deepening effect mentioned above, further raising the augmented innovation productivity, $\tilde{\psi} = (\mathbf{N}\eta)^\alpha \frac{\Psi}{N}$,

and, in turn, the composite parameter $\mathbf{Q}$.¹² In this case, the DPP efficiency affects the endogenous adoption of innovation technology through its impact on the relative prices of R&D inputs.

The three threshold values, $\tilde{\mathbf{Q}}^{me}$, $\tilde{\mathbf{Q}}^{sp}$, and $\hat{\mathbf{Q}}$, are independent of α , as shown in equation (26). Given $\frac{1}{N\eta} < 1$, the endogenous technology adoption amplifies data deepening and raises $\mathbf{Q}$, leading to a higher BGP growth rate under both allocations. Moreover, if this pushes $\mathbf{Q}$ above the threshold $\hat{\mathbf{Q}}$, the market equilibrium features a higher, but socially insufficient, BGP growth rate.¹³

3.3 Restoring Social Optimum in the Market Equilibrium

In this subsection, we examine two policy instruments that align the innovation incentives in the market equilibrium with the social optimum.

3.3.1 Patent Breath

Following Iwaisako and Futagami (2013), we assume that the markup ratio is given by $\mu = \beta z$, jointly determined by the quality step size z and patent breadth $\beta \in (\frac{1}{z}, 1]$. The case of $\mu = z$, analyzed in the previous subsections and shown in figure 1, corresponds to the full patent breadth ($\beta = 1$). A marginal reduction in β below unity lowers the markup ratio to $\mu = \beta z < z$, which in turn reduces the market-equilibrium BGP growth rate.¹⁴

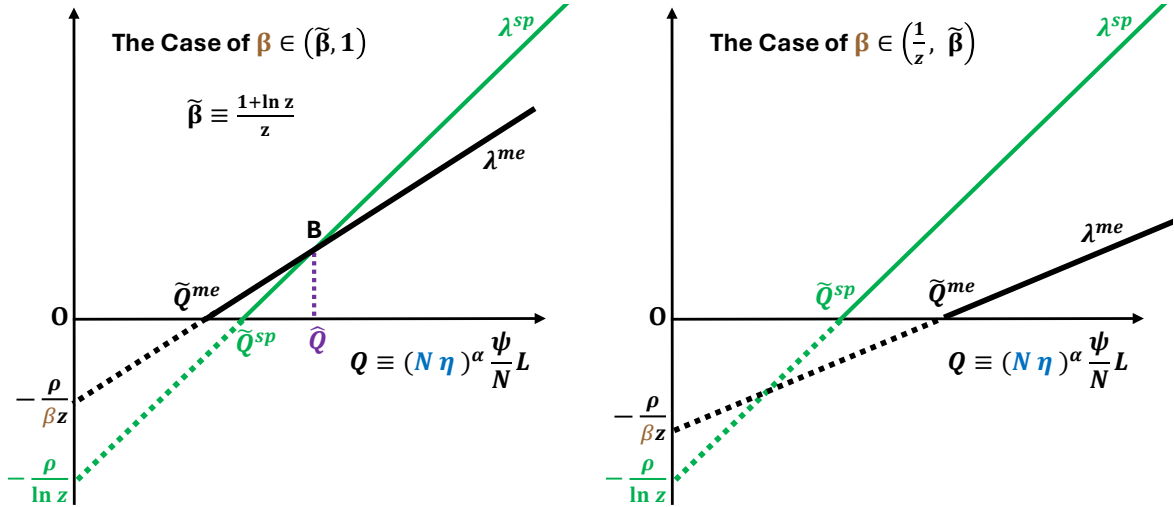


Figure 2: BGP Innovation Schedules under the Two Allocations: $\mu = \beta z < z$

¹²By contrast, if the DPP efficiency is so low that the normalized data bundle is more expensive than labor, i.e., $\frac{1}{N\eta} > 1$, innovators prefer the innovation technology with a lower data share and shift R&D expenditure toward labor. In that case, the endogenous technology adoption helps mitigate the relative-price effect on data use and $\tilde{\psi}$.

¹³Appendix 6.1 analyzes the case in which the arrival rate of innovation is a CES aggregator of the two R&D inputs. Suppose innovators can choose among alternative innovation technologies that differ in the elasticity of substitution. If the DPP efficiency is sufficiently high so that the data bundle is relatively cheaper than labor, i.e., $\frac{1}{N\eta} < 1$, innovators optimally adopt the innovation technology with a higher elasticity of substitution, which allows them to tilt input demand toward the relatively cheaper input, i.e., the data bundle. The cost-saving effect further amplifies data deepening, thereby raising the augmented innovation productivity.

¹⁴A lower markup reduces monopoly profits and thus weakens innovators' expected marginal return to R&D. Besides, by raising the wage rate $w_t = \frac{Z_t}{\mu}$, it increases the marginal cost of effective R&D time input.

In figure 2, the green lines show the social planner's BGP innovation schedules, identical to that in figure 1.¹⁵ When $\beta < 1$, the market-equilibrium schedule, represented by the black line, features a lower intercept, $-\frac{\rho}{\beta z}$, and a flatter slope, $(1 - \frac{1}{\beta z})$, than its counterpart in figure 1.

The left panel shows the case of a sufficiently large patent breadth, $\beta \in (\tilde{\beta}, 1)$, where $\tilde{\beta} \equiv \frac{1+\ln z}{z}$.¹⁶ In this case, the two innovation schedules intersect at point B, and the three threshold values of the composite parameter have the same ranking as in figure 1, $\tilde{Q}^{me} < \tilde{Q}^{sp} < \hat{Q}$. For $Q \in (\tilde{Q}^{me}, \hat{Q})$, the market-equilibrium BGP innovation rate exceeds the socially optimal level; for $Q > \hat{Q}$, the market equilibrium features socially insufficient innovation.

The right panel shows the case of a sufficiently low patent breadth, $\beta \in (\frac{1}{z}, \tilde{\beta})$, in which the two innovation schedules no longer intersect, and the ranking of the two threshold values reverses, $\tilde{Q}^{sp} < \tilde{Q}^{me}$. In this case, the market-equilibrium innovation rate is socially insufficient.

Based on the analysis above, patent breadth can be used as a policy instrument to restore the socially optimal BGP innovation rate in the market equilibrium, if the composite parameter lies between the two threshold values, $Q \in (\tilde{Q}^{sp}, \hat{Q})$, or equivalently if the DPP efficiency satisfies $\eta \in (\tilde{\eta}^{sp}, \hat{\eta})$, given $\mu = z$. In this case, the market equilibrium features socially excessive innovation, and restoring the social optimum requires setting patent breadth equal to¹⁷

$$\beta^* = \left(\frac{Q}{\rho} + 1\right) \frac{\ln z}{z} < 1, \text{ where } Q < \hat{Q}_z \equiv \rho \left(\frac{z}{\ln z} - 1\right). \quad (30)$$

Compared to the initial case of $\mu = z$, reducing the patent breadth below unity implies a lower markup ratio, $\mu = \beta^* z < z$, which reduces monopoly profits and raises the wage rate, thereby equalizing the market-equilibrium BGP innovation rate with the socially optimal rate. The higher the DPP efficiency, the higher the composite parameter Q , the lower the initial gap between the two BGP growth rates, the closer the optimal patent breadth with its upper bound of unity.

Given $\mu = z$, if $\eta > \hat{\eta}$ such that $Q > \hat{Q}$, the market-equilibrium innovation rate becomes socially insufficient, as shown in figure 1. Since patent breadth is already at its upper bound, $\beta = 1$, it cannot be increased further to restore social optimum in the market equilibrium.

3.3.2 Proportional Taxes and Subsidies

In this subsection, we keep patent breadth and the markup ratio fixed. For $Q < \hat{Q}$, the market-equilibrium BGP innovation rate exceeds the socially optimal level. As shown in equation (17) and explained in subsection 2.8, the innovator's input ratio is identical under the two allocations, i.e., $\frac{\tilde{D}_l}{s_{i,t}} = \frac{\alpha}{1-\alpha} N \eta$. Thus, restoring the socially optimal allocation in the market equilibrium requires applying the same tax rate to both inputs. Using the equivalent representation of the innovation function, it is sufficient to focus on the innovator's effective R&D time input and to choose an

¹⁵As shown in subsections 2.8 and 3.1, the innovation step size, z , which captures the social gains from innovation, plays a key role in the social planner's allocation, while prices do not enter the social planner's problem.

¹⁶We derive $\tilde{\beta}$ by equating the two threshold values, $\tilde{Q}^{me} = \frac{\rho}{\beta z - 1}$ and $\tilde{Q}^{sp} = \frac{\rho}{\ln z}$.

¹⁷The optimal patent breadth β^* is obtained by equating the BGP growth rates under the market equilibrium and the social planner's allocation, as characterized by equations (23) and (25).

appropriate tax rate τ . The innovator's first-order condition is¹⁸

$$\frac{(1 - \frac{1}{\mu}) \frac{Z_t m_t}{N}}{\rho + \lambda_t} N \tilde{\psi} = (1 + \tau) w_t = (1 + \tau) \frac{Z_t}{\mu}, \Rightarrow \lambda^{me}(\tau) = \frac{(1 - \frac{1}{\mu}) \mathbf{Q} - \frac{\rho}{\mu} (1 + \tau)}{1 + \frac{\tau}{\mu}} \quad (31)$$

Solve for the optimal tax rate τ^* that aligns the market-equilibrium BGP innovation rate with that under the social planner's allocation, as characterized by equations (31) and (25).

$$\lambda^{me}(\tau) = \lambda^{sp} \Rightarrow \tau^* = \frac{\hat{\mathbf{Q}} - \mathbf{Q}}{\rho + \mathbf{Q} - \tilde{\mathbf{Q}}^{sp}} \text{ where } \hat{\mathbf{Q}} \equiv \rho \left(\frac{\mu}{\ln z} - 1 \right) \text{ and } \tilde{\mathbf{Q}}^{sp} \equiv \frac{\rho}{\ln z}. \quad (32)$$

Proposition 1. For $\eta \in (\tilde{\eta}^{me}, \hat{\eta})$, the market equilibrium features a socially excessive BGP innovation rate, while restoring the socially optimal rate requires taxing the data bundle and R&D labor, $\tau^* > 0$.

For $\eta > \hat{\eta}$, the market equilibrium features a socially insufficient BGP innovation rate, while restoring the socially optimal rate requires subsidizing the data bundle and R&D labor, $\tau^* < 0$.

4 Endogenous Efficiency of Digital Privacy Protection

So far, the DPP efficiency, η , has been treated as exogenous. We endogenize η by allowing households and the data intermediary to undertake effort that enhances DPP efficiency. We then characterize how technological and regulatory factors shape their effort choices and analyze the resulting implications for data prices, the augmented innovation productivity, and long-run growth.

4.1 Household Efforts on Digital Literacy

Suppose individual households can devote $R_H \frac{d_t}{Z_t}$ units of time to improving their digital literacy¹⁹ and thereby enhance their DPP efficiency as follows,

$$\eta = \underline{\eta} (1 + \gamma_H R_H), \quad (33)$$

where $\underline{\eta}$ denotes the baseline DPP efficiency and γ_H measures the effectiveness of household effort in enhancing the DPP efficiency.²⁰ Thus, for each unit of the (normalized) data sales, $\frac{d_t}{Z_t}$, households need to spend R_H units of time on digital literacy and $\frac{1}{\underline{\eta}}$ units of time on DPP.

Households choose the effort ratio R_H to minimize their marginal time cost of data sales.

$$MTC_H \equiv R_H + \frac{1}{\underline{\eta} (1 + \gamma_H R_H)} \text{ and } \frac{\partial MTC_H}{\partial R_H} = 0, \Rightarrow R_H^* = \frac{1}{\gamma_H} \left(\sqrt{\frac{\gamma_H}{\underline{\eta}}} - 1 \right). \quad (34)$$

¹⁸The associated tax revenue, $\tau w_t s_{i,t}$, is rebated to households in a lump-sum fashion.

¹⁹It refers to households' efforts to acquire and update the knowledge and skills required to use digital tools safely, recognize online risks, make informed decisions about data sharing and protection, address the consequences of data leakage, and etc. These knowledge and skills must be updated continuously because digital threats and privacy-protection technologies evolve over time.

²⁰A higher γ_H reflects lower learning barriers or greater accessibility of the DPP-related knowledge and tools, e.g., through free and user-friendly online learning platforms. In the previous sections, we implicitly impose $\gamma_H = 0$, so that DPP efficiency is exogenous at the baseline level, $\eta = \underline{\eta}$.

Households chooses a positive effort ratio $R_H^* > 0$, if the effectiveness of their effort exceeds the baseline DPP efficiency, $\gamma_H > \underline{\eta}$. Otherwise, if $\gamma_H \leq \underline{\eta}$, the household chooses $R_H^* = 0$.

In the left panel of Figure 3, the downward-sloping, dashed curve shows the inverse of DPP efficiency as a decreasing function of the household effort ratio R_H , while the U-shaped solid curve shows the household marginal time cost, with the minimum at $R_H = R_H^*$, in the case of $\gamma_H = 3\underline{\eta}$.

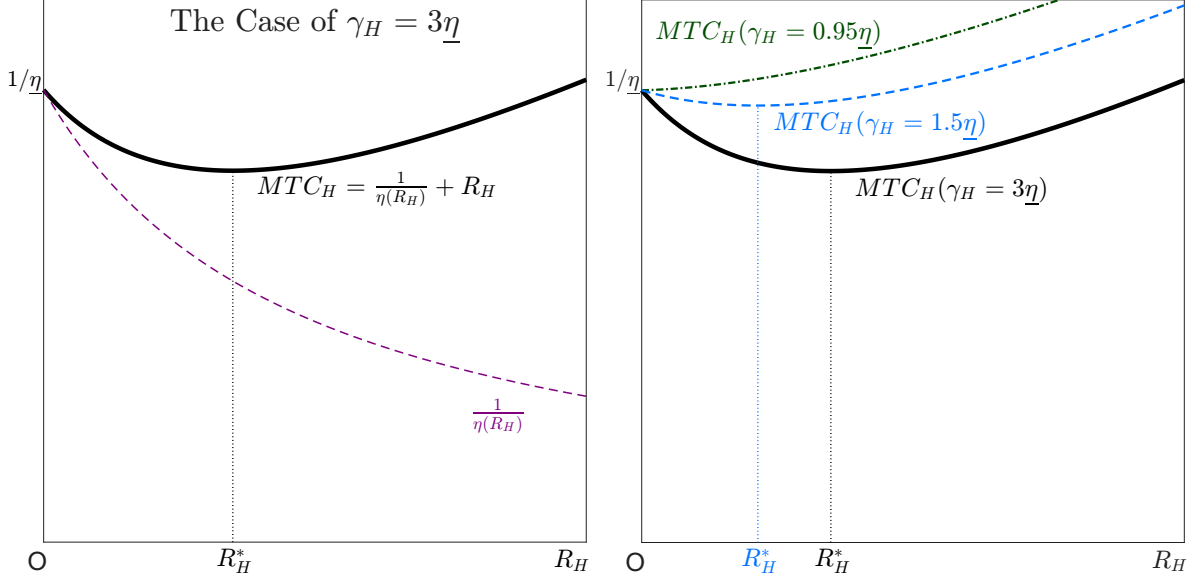


Figure 3: Household's Effort Ratio and Marginal Time Cost of Data Sales

The right panel plots the household's marginal time cost as a function of R_H for three values of the effectiveness parameter, $\gamma_H \in \{0.95\underline{\eta}, 1.5\underline{\eta}, 3\underline{\eta}\}$. Relative to the case of $\gamma_H = 3\underline{\eta}$, a lower effectiveness such as $\gamma_H = 1.5\underline{\eta}$ still generates a U-shaped pattern, as shown by the dashed curve, while the minimum is attained at a smaller optimal effort ratio R_H^* . If the effective parameter falls below the baseline DPP efficiency, e.g., $\gamma_H = 0.95\underline{\eta}$, the dash-dotted curve shows that the marginal time cost MTC_H increases monotonically for all $R_H > 0$. In this case, households choose $R_H^* = 0$.

In Figure 4, the dashed curve shows the household's optimal effort ratio R_H^* as a function of the effectiveness parameter γ_H .²¹ The dash-dotted curve and the solid curve show, respectively, the inverse of the DPP efficiency and the household's marginal time cost of data sales, both evaluated at the optimal effort choice R_H^* and both decreasing in γ_H .

Now, consider the social planner's allocation. The total time endowment of households is allocated across four activities: production of intermediate goods, R&D activities, DPP, and household effort on digital literacy.

$$m_t + s_t = L - R_H \frac{D_t}{Z_t} - \frac{1}{\eta} \frac{D_t}{Z_t} = L - MTC_S \cdot \tilde{D}_t, \text{ where } MTC_S \equiv R_H + \frac{1}{\underline{\eta}(1 + \gamma_H R_H)}. \quad (35)$$

denotes the social planner's marginal time cost of the (normalized) data bundle. The social planner chooses R_H to minimize MTC_S . As MTC_S takes the same function form as MTC_H , the social planner's optimal choice, R_H^{sp} , coincides with that of households, R_H^* ; see equation (34).

²¹The optimal effort ratio R_H^* reaches its maximum value $\frac{1}{4\underline{\eta}}$ at $\gamma_H = 4\underline{\eta}$. Beyond this threshold, further increases in γ_H reduce the household's optimal effort ratio.

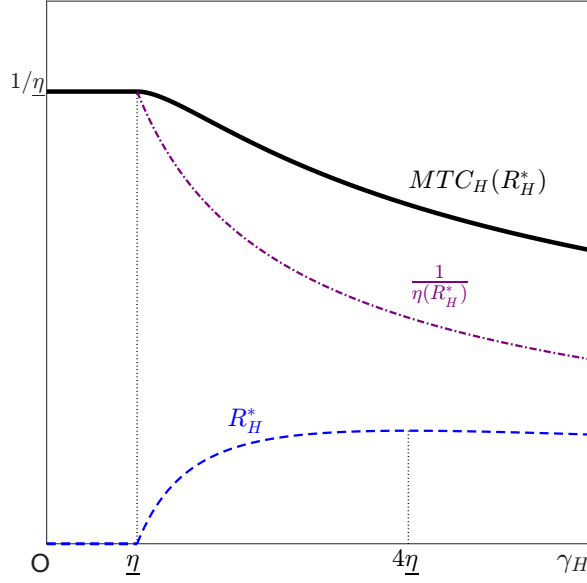


Figure 4: Impacts of γ_H on Household Marginal Time Cost

Define $\tilde{\eta}$ as the inverse of this marginal time cost. Substituting the household's optimal effort ratio R_H^* into the marginal time cost yields an expression for $\tilde{\eta}$ that is strictly increasing in the effectiveness parameter γ_H for all $\gamma_H > \underline{\eta}$:

$$\tilde{\eta} \equiv \frac{1}{\frac{1}{\eta(1+\gamma_H R_H^*)} + R_H^*} = \begin{cases} \frac{1}{\frac{2}{\sqrt{\gamma_H \underline{\eta}} - \frac{1}{\gamma_H}}} > \underline{\eta}, & \text{if } \gamma_H > \underline{\eta}, \\ \underline{\eta} & \text{if } \gamma_H \leq \underline{\eta}. \end{cases} \quad (36)$$

Intuitively, if DPP-related knowledge and skills become more readily available and accessible, households prefer to devote more time to improving digital literacy, which reduces their marginal time cost of data sales and raises $\tilde{\eta}$. As a result, households are willing to supply consumption data at a lower price, $p_{s,t} = \frac{1}{\tilde{\eta}} \frac{w_t}{Z_t}$, which in turn allows the intermediary to charge a lower price for the data bundle, $p_{b,t} = \frac{1}{\tilde{\eta} N} \frac{w_t}{Z_t}$. Following the approach in subsection 2.7, we can rewrite the Cobb–Douglas innovation technology in an equivalent representation,

$$\lambda_t = \lambda_{i,t} = \tilde{\psi} s_t, \text{ where } \tilde{\psi} \equiv \psi \frac{(N \tilde{\eta})^\alpha}{N}, \text{ and } s_t \equiv s_t + \frac{\tilde{D}_t}{\tilde{\eta}} = \frac{s_t}{1 - \alpha} \quad (37)$$

We can still use Figure 1 to show the BGP innovation schedules under the two allocations, with the only modification that the efficiency parameter η embedded in the composite parameter $\mathbf{Q}$ on the horizontal axis is now replaced by $\tilde{\eta}$. An increase in the effectiveness of households' effort, γ_H , lowers the marginal time cost of data sales, leading to a higher $\tilde{\eta}$ and, in turn, raising the augmented innovation productivity $\tilde{\psi}$. Through this channel, the composite parameter $\mathbf{Q}$ increases, leading to a higher BGP growth rate under both allocations, as shown in the left panel of Figure 5.

So far, we have assumed that households devote time to improving digital literacy, thereby raising the DPP efficiency. Under this assumption, the social planner's marginal time cost of data sharing coincides with that faced by households, $MTS_s = MTS_H$, so that the optimal effort ratio is identical under the two allocations, $R_H^* = R_H^{sp}$. As a result, no policy intervention is required.

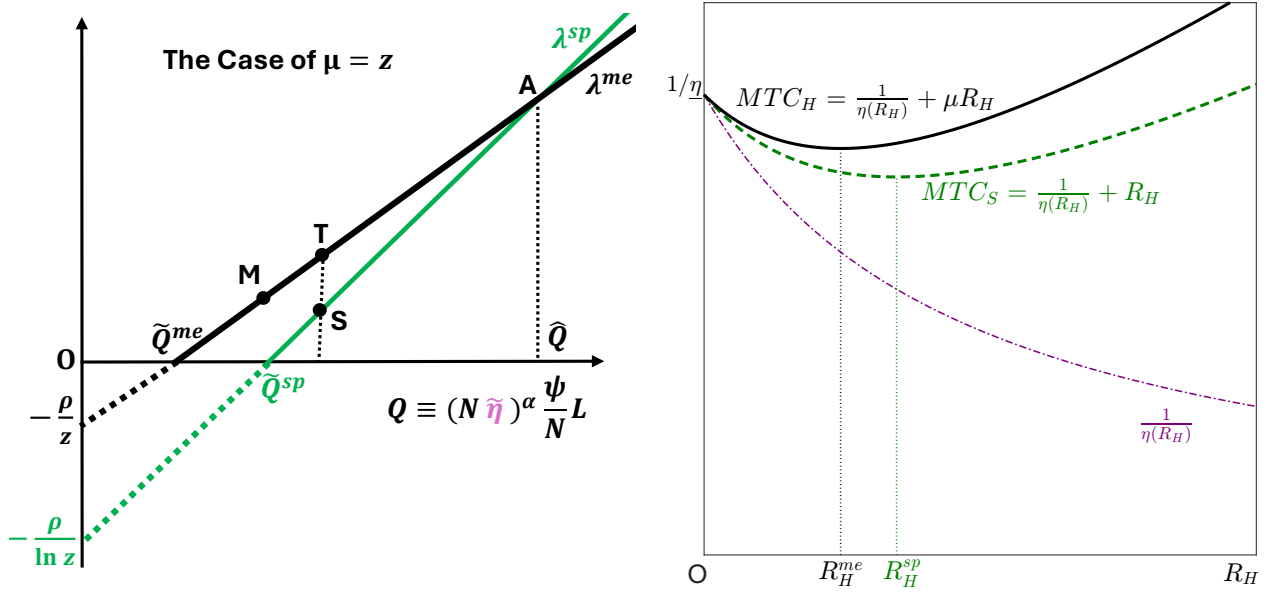


Figure 5: Household Input into Digital Literacy and Endogenous Marginal Time Cost of Data Sales

Let us consider an alternative specification in which households can improve digital literacy by spending $R_H d_t = R_H Z_t \frac{d_t}{Z_t}$ units of final goods on relevant products and services rather than using their time as the input. In this case, the household digital literacy spending per unit of the normalized data sales is $R_H Z_t$, and the DPP efficiency increases with R_H in the same manner as before, with $\eta \equiv \underline{\eta}(1 + \gamma_H R_H)$. In the market equilibrium, this expenditure can be converted into time units using the wage rate, $w_t = \frac{Z_t}{\mu}$, when computing the household's marginal time cost of data sharing. By contrast, in the social planner's allocation, the same expenditure is converted into time units using the marginal product of labor, $\frac{\partial y_t}{\partial m_t} = Z_t$, when computing the social planner's marginal time cost. In each allocation, R_H is chosen to minimize the marginal time cost,

$$MTC_H \equiv \frac{R_H Z_t}{w_t} + \frac{1}{\underline{\eta}(1 + \gamma_H R_H)} = \mu R_H + \frac{1}{\underline{\eta}(1 + \gamma_H R_H)}, \Rightarrow R_H^{me} = \frac{1}{\gamma_H} \left(\sqrt{\frac{\gamma_H}{\mu \underline{\eta}}} - 1 \right), \quad (38)$$

$$MTC_S \equiv \frac{R_H Z_t}{Z_t} + \frac{1}{\underline{\eta}(1 + \gamma_H R_H)} = R_H + \frac{1}{\underline{\eta}(1 + \gamma_H R_H)}, \Rightarrow R_H^{sp} = \frac{1}{\gamma_H} \left(\sqrt{\frac{\gamma_H}{\underline{\eta}}} - 1 \right) \quad (39)$$

Due to monopolistic markup by industry leaders, the equilibrium wage rate is below the marginal product of labor, $w_t = \frac{Z_t}{\mu} < Z_t$. As shown in the right panel of Figure 5, this wedge, captured by $\mu > 1$, keeps the household's marginal time cost curve above that of the social planner. As a result, households choose a spending ratio below the socially efficient level, $R_H^{me} < R_H^{sp}$; see equations (38)–(39). The marginal time cost in the market equilibrium therefore exceeds its socially efficient counterpart, $MTC_H(R_H^{me}) > MTC_S(R_H^{sp})$. Restoring the socially optimal spending ratio in the market equilibrium requires a proportional subsidy on household spending in final goods, $(1 - \tau_H)R_H$, which offsets the markup wedge when we convert the expenditure into time units.

$$\frac{(1 - \tau_H^*)R_H Z_t}{w_t} = \frac{R_H Z_t}{Z_t}, \Rightarrow \tau_H^* = 1 - \frac{1}{\mu}. \quad (40)$$

The left panel of Figure 5 illustrates a particular example where point **S** shows the social planner's BGP innovation rate, with $\tilde{\eta}^{sp} = \frac{1}{MTC_S(R_H^{sp})}$, while point **M** shows the market-equilibrium BGP innovation rate, with $\tilde{\eta}^{me} = \frac{1}{MTC_H(R_H^{me})} < \tilde{\eta}^{sp}$. Implementing the social optimum in the market equilibrium requires two policy instruments in this example.

First, a proportional subsidy to household spending on digital literacy (in units of final goods) reduces the household's marginal time cost and raises $\tilde{\eta}^{me}$, thereby shifting the allocation from point **M** to point **T**. However, as the composite parameter at point **T** (**S**) is below the threshold value $\hat{Q}$, the resulting market-equilibrium innovation rate at point **T** remains socially excessive. Restoring social optimum in the market equilibrium requires a proportional tax on R&D inputs, as characterized in equation (32), which further shifts the allocation from point **T** to the socially efficient point **S**. In this example, subsidies to digital-literacy-related household spending coexist with taxes on R&D inputs (including the data bundle and labor), because each policy instrument corrects a distinct distortion.

Our analysis is subject to an important caveat. Unlike conventional market transactions, the sharing of consumption data typically delivers immediate, front-loaded benefits, e.g., discounts, access to free digital services, or personalized offerings, while generating back-loaded costs in the form of future risks of data breaches, leakage, discrimination, or manipulation. Because these costs materialize only with delay, they are often imperfectly internalized at the time of data provision, potentially due to bounded rationality, hyperbolic discounting, and related behavioral biases (Acquisti, Brandimarte, and Loewenstein, 2015; Acquisti, Taylor, and Wagman, 2016). When households' initial level of digital literacy is very low, a marginal improvement may raise their awareness of the risks associated with online activity and data sharing, thereby raising rather than reducing the marginal time cost of DPP. A formal analysis of this channel is left for future research.

4.2 The Intermediary's Input for Data Security

In addition to households' time input in digital literacy, we allow the data intermediary to hire $R_I \frac{D_t}{Z_t}$ units of labor to strengthen data security and reduce the risk of data breaches, thereby further improving the household DPP efficiency. For expositional simplicity, we assume that the DPP efficiency takes a multiplicative form,

$$\eta = \underline{\eta} x_H x_I, \quad \text{where } x_j = 1 + \gamma_j R_j \text{ and } j \in \{H, I\}. \quad (41)$$

The parameter γ_I captures the effectiveness of the intermediary's data-security-related hiring, which depends not only on the availability of privacy-enhancing technologies (Bains and Gaidosch, 2025; OECD, 2025), but also on the regulatory and policy frameworks (OECD, 2023; Demirci, Israeli, and Ascarza, 2025) that govern and enforce the intermediary's adoption of these technologies. In the following, we analyze how the two effectiveness parameters, γ_I and γ_H , jointly affect households' digital-literacy effort and the intermediary's data-security hiring decisions.

In the market equilibrium, the data intermediary purchases consumption data $d_t L$ from households at the price $p_{s,t} = MTS_H \cdot \frac{w_t}{Z_t}$, converts them into a data bundle $D_t = d_t L$ and sells identical

copies to N innovators at the price $p_{b,t}$, and hires $R_I \frac{D_t}{Z_t}$ units of labor at the wage rate w_t to enhance data security. Use the intermediary's zero-profit condition to solve for the price of the data bundle,

$$p_{s,t} d_t L + R_I \frac{D_t}{Z_t} w_t = N D_t p_{b,t}, \Rightarrow p_{b,t} = \left[\frac{1}{\underline{\eta} (1 + \gamma_H R_H^*) (1 + \gamma_I R_I)} + R_H^* + R_I \right] \frac{1}{N} \frac{w_t}{Z_t}, \quad (42)$$

where R_H^* denotes the household's optimal effort ratio. Due to competition from potential entrants, the intermediary chooses the hiring ratio R_I to minimize the price of the data bundle,²²

$$\frac{\partial p_{b,t}}{\partial R_I} = 0, \Rightarrow R_I^* = \frac{1}{\gamma_I} \left[\sqrt{\frac{\gamma_I}{\underline{\eta} (1 + \gamma_H R_H^*)}} - 1 \right]. \quad (43)$$

Taking the intermediary's optimal hiring ratio R_I^* as given, each household treats $\underline{\eta} (1 + \gamma_I R_I^*)$ just like the baseline DPP efficiency in subsection 4.1 and chooses the effort ratio R_H to minimize its marginal time cost of data sales. Use equation (34) to get the household optimal effort ratio

$$R_H^* = \frac{1}{\gamma_H} \left[\sqrt{\frac{\gamma_H}{\underline{\eta} (1 + \gamma_I R_I^*)}} - 1 \right]. \quad (44)$$

Combine equations (43) and (44) to solve for the two ratios as the functions of parameters.

$$R_H^* = \frac{1}{\gamma_H} \left[\left(\frac{\gamma_H^2}{\underline{\eta} \gamma_I} \right)^{\frac{1}{3}} - 1 \right] \quad \text{and} \quad R_I^* = \frac{1}{\gamma_I} \left[\left(\frac{\gamma_I^2}{\underline{\eta} \gamma_H} \right)^{\frac{1}{3}} - 1 \right]. \quad (45)$$

The left panel of figure 6 shows four regions in the (γ_I, γ_H) space.²³

- For the parameter pair of (γ_I, γ_H) in region **N**, both effectiveness parameters are below the baseline DPP efficiency, i.e., $\gamma_j < \underline{\eta}$ for $j \in \{H, I\}$. In this case, both households and the intermediary choose $R_H^* = R_I^* = 0$.
- For the parameter pair of (γ_I, γ_H) in region **H**, households choose a positive effort ratio, $R_H^* > 0$, while the intermediary chooses $R_I^* = 0$.
- For the parameter pair of (γ_I, γ_H) in region **I**, the intermediary chooses a positive hiring ratio, $R_I^* > 0$, while households choose $R_H^* = 0$.
- For the parameter pair of (γ_I, γ_H) in region **HI**, both households and the intermediary choose the positive ratios, with $R_H^* > 0$ and $R_I^* > 0$.

²²As shown in equation (52) and Appendix 6.2, each household takes the intermediary's hiring ratio R_I as given and choose its own effort ratio R_H to minimize its marginal time cost of data sales. Anticipating this behavior, the intermediary could in principle use R_I to influence households' effort choices and thereby reduce the price of the data bundle. However, according to the Envelope Theorem, households' optimal responses fully offset any indirect effect of the intermediary's decision on the price of the data bundle. Consequently, the intermediary chooses R_I , taking households' optimal behavior as given, and does not need to internalize its impact on the households' effort ratio.

²³The border between regions **I** and **HI** is characterized by $\gamma_H = \sqrt{\underline{\eta} \gamma_I}$, while the border between regions **H** and **HI** is given by $\gamma_H = \frac{\gamma_I^2}{\underline{\eta}}$.

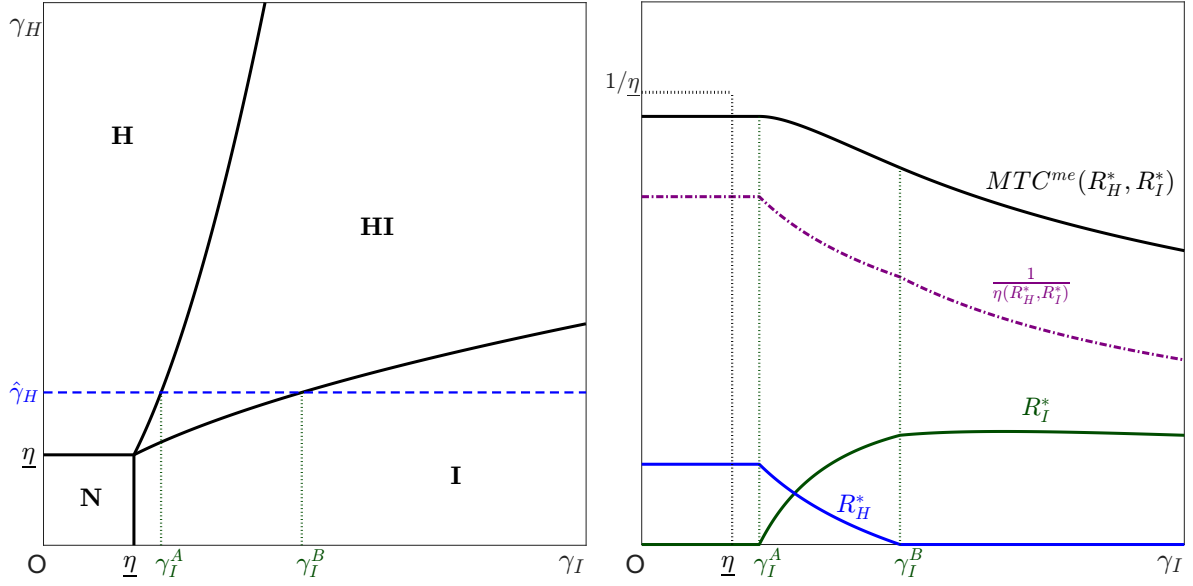


Figure 6: Joint Determination of the Effort Ratio and the Hiring Ratio

Consider the case $\gamma_H = \hat{\gamma}_H > \underline{\eta}$, as shown in the left panel of Figure 6. Suppose that technological and regulatory improvements make the intermediary's data-security-related hiring more effective, captured by a rise in γ_I from zero. There exist two threshold values, $\gamma_I^B > \gamma_I^A > \underline{\eta}$ splitting the parameter space into three intervals.

First, for $\gamma_I \in (0, \gamma_I^A)$, the parameter pair (γ_I, γ_H) lies in region **H**. As shown by the blue, horizontal line in the right panel, the household's optimal effort ratio remains constant at $R_H^* > 0$, while the intermediary optimally chooses not to hire labor for data security, i.e., $R_I^* = 0$.

Second, for $\gamma_I \in (\gamma_I^A, \gamma_I^B)$, the parameter pair lies in region **HI**. The upward-sloping, dark-green curve in the right panel shows that the intermediary's optimal hiring ratio R_I^* increases with γ_I , whereas the downward-sloping, blue curve shows that the household's optimal effort ratio R_H^* decreases with γ_I . This opposite co-movement reflects strategic substitution between household effort and intermediary hiring in enhancing DPP efficiency.

Finally, for $\gamma_I > \gamma_I^B$, the parameter pair lies in region **I**. In this case, households optimally exert no effort in digital literacy, $R_H^* = 0$, while the intermediary's data-security-related hiring ratio follows the pattern shown in Figure 4.

For expositional simplicity, we assume that households' digital-literacy effort and the intermediary's data-security hiring affect the DPP efficiency in a multiplicative manner, as shown in equation (41). This formulation implies a unit elasticity of substitution between x_H and x_I . Alternatively, one could assume that DPP efficiency is given by a constant-elasticity-of-substitution (CES) aggregator of x_H and x_I , as follows

$$\eta = \underline{\eta} \left[\beta \left(\frac{x_H}{\beta} \right)^{\frac{\sigma-1}{\sigma}} + (1-\beta) \left(\frac{x_I}{1-\beta} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \text{ where } x_j = 1 + \gamma_j R_j \text{ and } j \in \{H, I\}. \quad (46)$$

$\beta \in (0, 1)$ denotes the factor share of x_H , and $\sigma > 0$ is the elasticity of substitution between x_H and x_I . This specification allows for the analysis of cases in which household's effort and intermediary's hiring are substitutes in terms of their impacts on the DPP efficiency, i.e., $\sigma > 1$, or

complements, i.e., $\sigma \in (0, 1)$. Then, one can explore how changes in the effectiveness parameters affect the two ratios in these cases.

As shown in Appendix 6.2, the two ratios in the market equilibrium are socially optimal in the current setting, and no policy intervention is therefore required. As explained in subsection 4.1, if either households' input in digital literacy or the intermediary's input in data security takes the form of final goods, the wedge between the wage rate and the marginal product of labor, $\frac{w_t}{Z_t} = \frac{1}{\mu} < 1$, creates a gap between the marginal time cost of data sales in the market equilibrium and that faced by the social planner. In that case, a proportional subsidy on the relevant expenditure, $\tau = 1 - \frac{1}{\mu}$, is required to offset this wedge and restore the social optimum.

5 Final Remarks

We introduce the time cost of digital privacy protection in a Schumpeterian growth model and analyze its impact on the supply of consumption data and the cost of data-driven innovation. A high marginal time cost of digital privacy protection may result from a low level of digital literacy among households, insufficient data-security investment by data intermediaries, or weak regulatory and institutional frameworks, which raises households' privacy concerns and reduces their willingness to sell consumption data. As a result, the cost of data-driven R&D increases, dampening innovative activity and lowering long-run economic growth.

When this marginal time cost exceeds a threshold value, the market equilibrium features socially excessive innovation, due to the intertemporal business-stealing effect. Restoring the socially optimal allocation requires policy interventions, such as taxes on data and other R&D inputs or a reduction in patent breadth, which further reduce long-run growth.

If households' digital-literacy effort and the data intermediary's data-security input can effectively reduce this marginal time cost, the aggregate cost of data for R&D declines, thereby stimulating innovation and long-run growth. Technological and regulatory improvements that enhance the effectiveness of these efforts promote data-driven innovation indirectly through the data-cost channel. Depending on how these efforts jointly shape this marginal time cost and on their relative effectiveness, households' and the intermediary's decisions may exhibit strategic substitutability.

To highlight the core mechanism in a tractable manner, we have made several simplifying assumptions, leaving several extensions for future research. First, our core mechanism can be embedded in a semi-endogenous growth model to eliminate scale effects. Second, endogenizing the degree of digitization, κ , would allow an analysis of the technological and regulatory determinants of digital infrastructure development. Third, the assumption of free entry implies a zero-profit condition for the data intermediary. Relaxing this assumption by introducing alternative market structures in the data-intermediary sector would allow an examination of their implications for data pricing, market power, and optimal regulation (Bergemann and Bonatti, 2024).

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6 Appendix

6.1 The Case of the CES Innovation Function

Our model assumes a linear time cost of DPP and imposes the intermediary’s zero-profit condition, which implies that the price of the data bundle is proportional to the wage rate, $p_{b,t} = \frac{1}{\eta N} \frac{w_t}{Z_t}$.

For expositional simplicity, we assume that the arrival rate of innovation is a Cobb–Douglas function of the two R&D inputs. Combining the innovator’s first-order conditions with the proportionality of the wage rate versus the price of the data bundle, the innovator’s optimal input ratio is constant, as shown in equation (17). It allows us to rewrite the arrival rate of innovation as a linear function of the effective time input for R&D, s_t , and the relative price of the data bundle versus labor, $\frac{1}{N\eta}$, is featured in the augmented innovation productivity parameter,

$$\tilde{\psi} \equiv \frac{\psi}{N} \frac{1}{\left(\frac{1}{N\eta}\right)^\alpha} = \frac{\psi}{N} (N\eta)^\alpha. \quad (47)$$

Now, suppose that the arrival rate of innovation is a CES function of the two inputs, with $\alpha \in (0, 1)$ denoting the factor share of the data bundle and ε the elasticity of substitution between the two inputs.²⁴

$$\lambda_{i,t} = \psi \left[\alpha \left(\frac{\tilde{D}_t}{\alpha} \right)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\alpha) \left(\frac{s_{i,t}}{1-\alpha} \right)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (48)$$

Combine the innovator’s optimal demand for R&D inputs with the proportionality of the wage rate versus the price of the data bundle. Again, the data bundle is proportional to the R&D labor,

$$\frac{\tilde{D}_t}{\alpha} \frac{1}{(N\eta)^\varepsilon} = \frac{s_t}{1-\alpha}, \Rightarrow \tilde{D}_t = \frac{(N\eta)^\varepsilon}{N} \frac{\alpha}{1-\alpha} s_t. \quad (49)$$

Define the effective time input for R&D as the sum of labor and the time for DPP. The innovation function becomes linear in the effective time input for R&D, with the augmented innovation productivity as follows.

$$m_t = L - \frac{\tilde{D}_t}{\eta} - s_t = L - s_t, \text{ where } s_t \equiv \frac{\tilde{D}_t}{\eta} + s_t = [\alpha(N\eta)^{\varepsilon-1} + (1-\alpha)] \frac{s_t}{1-\alpha} \quad (50)$$

$$\lambda_t = \lambda_{i,t} = \tilde{\psi} s_t, \text{ where } \tilde{\psi} \equiv \frac{\psi [\alpha(N\eta)^{\varepsilon-1} + (1-\alpha)]^{\frac{1}{\varepsilon-1}}}{N} = \frac{\psi}{N} \frac{1}{[\alpha(\frac{1}{N\eta})^{1-\varepsilon} + (1-\alpha)]^{\frac{1}{1-\varepsilon}}}. \quad (51)$$

²⁴As $\varepsilon \rightarrow 1$, the innovation function (48) converges to the Cobb–Douglas form specified in equation (11).

Switching the innovation function from the Cobb–Douglas to the CES form does not affect the three threshold values, $\tilde{\mathbf{Q}}^{me}$, $\tilde{\mathbf{Q}}^{sp}$, and $\hat{\mathbf{Q}}$, as defined in equation (26).

When $\varepsilon > 1$, labor and the data bundle are substitutes in R&D. If DPP efficiency is sufficiently high so that the data bundle is cheaper than labor, i.e., $\frac{1}{\eta N} < 1$, innovators purchase more data bundle relative to labor than in the Cobb–Douglas case; see equation (49). This cost-saving reallocation raises the augmented innovation productivity $\tilde{\psi}$, making the composite parameter $\mathbf{Q} = \tilde{\psi}L$ higher than in the Cobb–Douglas case. The larger the ε , the stronger this substitution effect, the more likely that $\mathbf{Q}$ exceeds the threshold $\hat{\mathbf{Q}}$. If so, the market equilibrium features a higher BGP innovation rate, which is socially insufficient; see Figure 1.

When $\varepsilon < 1$, labor and the data bundle are complements in R&D. In this case, if the data bundle is relatively cheaper than labor, i.e., $\frac{1}{\eta N} < 1$, the augmented innovation productivity parameter falls below its Cobb–Douglas counterpart. As a result, the market equilibrium features a lower BGP innovation rate, which is more likely to be socially excessive.

To sum up, the DPP efficiency η , by affecting the relative cost of the (normalized) data bundle in labor units, $\frac{1}{\eta N}$, plays a central role in our model. Whether the market equilibrium features socially excessive or insufficient innovation depends on the factor share α and the elasticity of substitution ε .

6.2 Joint Determination of the Two Ratios

Suppose the DPP efficiency is an increasing function of the household’s digital-literacy effort ratio, R_H , and the intermediary’s data-security hiring ratio, R_I . Define $x_j \equiv 1 + \gamma_j R_j$ as a linear proxy of R_j , where $j \in \{H, I\}$. As shown below, this substitution simplifies the technical derivation.

$$\eta = \underline{\eta} x_H x_I \quad \text{and} \quad \frac{\partial \ln \eta}{\partial \ln x_j} = 1.$$

Individual households take x_I as given²⁵ and choose x_H to minimize their marginal time cost.²⁶

$$\min_{x_H} MTC_H \equiv R_H + \frac{1}{\eta} = \frac{x_H - 1}{\gamma_H} + \frac{1}{\eta}, \Rightarrow \frac{1}{\gamma_H} - \frac{1}{\eta} \frac{1}{x_H} \frac{\partial \ln \eta}{\partial \ln x_H} = 0, \Rightarrow x_H^* = \sqrt{\frac{\gamma_H}{\underline{\eta} x_I}}. \quad (52)$$

The household’s optimal choice of x_H depends on the intermediary’s choice x_I . Anticipating this response, the intermediary could in principle choose x_I to influence the household’s choice of x_H and thereby minimize the price of the data bundle; see equation (42).

$$\begin{aligned} \min_{x_I} R_H + R_I + \frac{1}{\eta} &= \frac{x_H - 1}{\gamma_H} + \frac{x_I - 1}{\gamma_I} + \frac{1}{\eta}, \\ \Rightarrow \frac{1}{\gamma_H} \frac{x_H}{x_I} \frac{\partial \ln x_H}{\partial \ln x_I} + \frac{1}{\gamma_I} - \frac{1}{\eta} \frac{1}{x_I} \left(\frac{\partial \ln \eta}{\partial \ln x_H} \frac{\partial \ln x_H}{\partial \ln x_I} + \frac{\partial \ln \eta}{\partial \ln x_I} \right) &= 0 \\ \left(\frac{1}{\gamma_H} - \frac{1}{\eta} \frac{1}{x_H} \frac{\partial \ln \eta}{\partial \ln x_H} \right) \frac{x_H}{x_I} \frac{\partial \ln x_H}{\partial \ln x_I} &= \frac{1}{\eta} \frac{1}{x_I} \frac{\partial \ln \eta}{\partial \ln x_I} - \frac{1}{\gamma_I}. \end{aligned}$$

²⁵Each household recognizes that its individual choice cannot influence the intermediary’s decision.

²⁶Since x_j is linear in R_j , choosing x_j is equivalent to choosing R_j , which simplifies the derivation.

The term in the brackets on the left-hand side is identical as the household's first-order condition with respect to R_H ; see equation (52). Thus, the intermediary's first-order condition with respect to R_I is simplified as

$$\frac{1}{\eta} \frac{1}{x_I} \frac{\partial \ln \eta}{\partial \ln x_I} - \frac{1}{\gamma_I} = 0, \Rightarrow x_I^* = \sqrt{\frac{\gamma_I}{\eta x_H^*}}. \quad (53)$$

Combine equations (52) and (53) to solve for the two ratios in the market equilibrium,

$$R_H^{me} = \frac{1}{\gamma_H} \left[\left(\frac{\gamma_H^2}{\eta \gamma_I} \right)^{\frac{1}{3}} - 1 \right] \quad \text{and} \quad R_I^{me} = \frac{1}{\gamma_I} \left[\left(\frac{\gamma_I^2}{\eta \gamma_H} \right)^{\frac{1}{3}} - 1 \right]. \quad (54)$$

Households aim to minimize the marginal time cost of data sales, an objective consistent with that of the intermediary. By the Envelope Theorem, the intermediary does not benefit from indirectly manipulating households' choices. Thus, the equilibrium is equivalent to one in which households and the intermediary take each other's actions as given and independently solve their respective optimization problems.

Now consider the social planner's allocation. The household's total time endowment is allocated across five uses: (i) production of intermediate goods, (ii) R&D, (iii) DPP, (iv) household effort to improve digital literacy, and (v) the intermediary's labor hiring for data security.

$$m_t + s_t = L - R_H \frac{d_t L}{Z_t} - R_I \frac{D_t}{Z_t} - \frac{1}{\eta} \frac{d_t L}{Z_t} = L - MTC_S \cdot \tilde{D}_t, \quad \text{where} \quad MTC_S \equiv R_H + R_I + \frac{1}{\eta}. \quad (55)$$

denotes the social planner's marginal time cost of the (normalized) data bundle. The social planner chooses, respectively, R_H and R_I to minimize its marginal time cost.

$$\min_{x_H, x_I}, MTC_S \equiv \frac{x_H - 1}{\gamma_H} + \frac{x_I - 1}{\gamma_I} + \frac{1}{\eta}, \Rightarrow \frac{1}{\gamma_j} - \frac{1}{\eta} \frac{1}{x_j} \frac{\partial \ln \eta}{\partial \ln x_j} = 0, \quad (56)$$

$$\Rightarrow x_j^{sp} = \sqrt{\frac{\gamma_j}{\eta x_k^{sp}}}, \quad \text{and} \quad R_j^{sp} = \frac{1}{\gamma_j} \left[\left(\frac{\gamma_j^2}{\eta \gamma_k} \right)^{\frac{1}{3}} - 1 \right], \quad \text{where } j, k \in \{H, I\} \text{ and } j \neq k. \quad (57)$$

As the solutions to household's effort ratio and the intermediary's hiring ratio are identical under the two allocation, no policy intervention is required in this case.